Parishram 2027

Chemistry

Solutions

DPP: 1

Q1 If 2 moles of $CaCO_3$ is dissolved in 900 g of water then percentage by mass of solution formed is

(A) 36.36% (B) 9.9%

(C) 18.18% (D) 11.75%

Q2 A solute A was put into a solvent to make a solution of mass 240 gram. What should be the mass of A so that its mass $\%$ in solution is 40 ?

(A) $96g$

(B) 24 g

(C) $48g$

(D) 36 g

Q3 A solution is made by mixing 50 g of water 20 g of salt and 30 g of sugar. Mass % of salt and sugar are respectively.

(A) 10,40 (B) 20,40

(C) 20,30 (D) 50,50

Q4 Find mass by mass % of a solution of 20 ml volume having solute of mass 4 g. (density of solution $= 1.4$ g/ml)

(A) 7.14% (B) 8.2%

(C) 9.64% (D) 14.28%

Q5 A tank contains 5 moles of oxygen, 2 moles of nitrogen and 20 g of hydrogen at room temperature.

Find the mole fraction of hydrogen.

(A) $\frac{5}{17}$

(B) $\frac{2}{15}$

(C) $\frac{8}{15}$

(D) $\frac{10}{17}$

Q6 An aqueous solution of ethanol contains 23 g of ethanol dissolved in 90 g of water. Find mole fraction of ethanol in the solution.

(A) $\frac{3}{4}$

(B) $\frac{3}{5}$

(C) $\frac{2}{7}$

(D) $\frac{1}{11}$

Q7 Which of the following has no units?

(A) Molarity (B) Normality

(C) Molality (D) Mole fraction

Q8 What is the number of ppm of $NaCl$ in the solution with 117 g of $NaCl$ dissolved in 500 mL of water?

(A) 1.89×10^5

(B) 2.78×10^5

(C) 9.12×10^5

(D) 6.54×10^5

Q9 Find molarity of a sample of pure water. Consider density $= 1$ g/ml.

(A) 27.7 (B) 33.3

(C) 55.5 (D) 11.1

Q10 Find out the molarity of 93% (w/W) H_2SO_4 (density $= 1.84$ g/ml).

(A) 174.6M

(B) 17.46M

(C) 1.746M

(D) All of these

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Answer Key

Q1	(C)
Q2	(A)
Q3	(C)
Q4	(D)
Q5	(D)
Q6	(D)
Q7	(D)
Q8	(A)
Q9	(C)
Q10	(B)


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